

Determination of the Rydberg Constant from the Hydrogen Balmer Series

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Abstract

The Rydberg constant was determined from the Balmer series of hydrogen using a spectrometer and Gaussian fitting to extract emission wavelengths. A linearized Rydberg relation yielded

$$R_H = 10975956.87 \pm 125.91 \text{ m}^{-1}$$

which differs from the accepted CODATA value by only 0.075%. The results demonstrate high precision, with deviations likely due to systematic effects such as refractive index and peak-shape modeling.

Introduction

The Rydberg constant is a fundamental parameter in atomic physics that characterizes the hydrogen emission spectrum. In this experiment, the constant was determined by analyzing the Balmer series, specifically the H- α , H- β , H- γ , and H- δ lines emitted by a hydrogen discharge lamp. The spectral lines were recorded using a spectrometer, and Gaussian fitting was applied to the data to identify peak emission wavelengths with high precision. The measured wavelengths were then used in the linearized Rydberg equation to calculate an experimental value of the constant, which was subsequently compared with the accepted CODATA value.

Aim of This Experiment

The aim of this investigation was to determine the Rydberg constant (R_H), by measuring the wavelengths ($\lambda_{\text{measured}}$) of several Balmer series emission lines in the visible spectrum of hydrogen.

Relevant Theory

The Balmer Series and the Rydberg Formula

The visible emission spectrum of hydrogen, known as the Balmer series, arises from electron transitions from higher energy levels ($n_i > 2$) to the $n = 2$ level. The wavelength λ of each line is given by the Rydberg formula:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{4} - \frac{1}{n_i^2} \right) \quad (1)$$

where:

- λ is the emission wavelength (m),
- R_H is the Rydberg constant for hydrogen (m^{-1}),
- n_i is the principal quantum number of the upper level ($n_i = 3, 4, 5, \dots$).

The first four Balmer lines are denoted as:

- H- α (656.3 nm, $n_i = 3$),
- H- β (486.1 nm, $n_i = 4$),
- H- γ (434.0 nm, $n_i = 5$), and
- H- δ (410.2 nm, $n_i = 6$).

In this investigation, these four lines were analyzed to determine the experimental value of the Rydberg constant.

Equation (1) was linearized to allow for experimental determination of R_H . By defining:

$$y = \frac{1}{\lambda}, \quad x = \left(\frac{1}{4} - \frac{1}{n_i^2} \right),$$

the equation becomes:

$$y = R_H \cdot x \quad (2)$$

A plot of y versus x yields a straight line, with slope equal to the Rydberg constant. The slope was obtained using a weighted linear least-squares fit.

Gaussian Fit for Wavelength Determination

Measured spectral peaks are broadened by instrumental effects. To determine the central emission wavelength λ_0 of each line with greater accuracy, a Gaussian function was fitted to the measured counts data.:

$$I(\lambda) = a \cdot e^{\left(-\frac{(\lambda - \lambda_0)^2}{2\sigma^2}\right)} \quad (3)$$

where:

- $I(\lambda)$ is the number of counts at wavelength λ ,
- a is the peak amplitude,
- λ_0 is the central wavelength (the parameter of interest),
- σ is the standard deviation, defining the peak width.

The fitting algorithm optimized parameters ' a ', λ_0 , and σ to best match the experimental data. The extracted λ_0 for each Balmer line was then used in the Rydberg equation (2). This approach provided a more reliable estimate than simply identifying the raw maximum intensity, as it accounted for the peak shape and reduced the influence of noise.

Methodology

Apparatus

The main apparatus used in this investigation was an optical spectrometer system (Heath EU-700 Czerny–Turner monochromator with a photomultiplier detector and pulse-counting electronics). The spectrometer was powered by an AC source, and both systems were connected to an oscilloscope to visualize the signals. A *HeNe* laser was employed for calibration, while a hydrogen discharge lamp served as the source for the Balmer series measurements. A mirror and a lens were used to focus the light from the lamp onto the entrance slit of the spectrometer.

Measurements

Three sets of measurements were performed:

1. ***HeNe* laser spectrum** (for calibration),
2. **Hydrogen lamp spectrum** (Balmer lines for Rydberg constant determination),
3. **Background spectrum** (for noise subtraction).

All measurements were carried out in a darkened laboratory to minimize background light.

HeNe Laser Calibration

The *HeNe* laser was carefully aligned with the spectrometer slit. The spectrum was recorded with a slit width of 100 μm , dwell time of 300 ms, and a wavelength range of $\lambda \in [6303, 6370] \text{ \AA}$, in increments of 0.5 $\text{\AA}$. A corresponding background measurement was taken with identical settings but with the laser switched off.

Hydrogen Lamp (Balmer Series)

The Balmer lines were measured using a slit width of 150 μm , dwell time of 300 ms, and increments of 0.2 $\text{\AA}$. After each line measurement, a background spectrum was recorded over the same wavelength range. Prior to measurement, the hydrogen lamp was carefully focused onto the slit using the mirror–lens setup.

The wavelength ranges for each Balmer line were:

- H- α : $\lambda \in [6550, 6600] \text{ \AA}$,

- H- β : $\lambda \in [4862, 4895] \text{ \AA}$,
- H- γ : $\lambda \in [4340, 4370] \text{ \AA}$,
- H- δ : $\lambda \in [4340, 4370] \text{ \AA}$.

Noise Subtraction, Calibration, and Uncertainty Analysis

Noise Subtraction

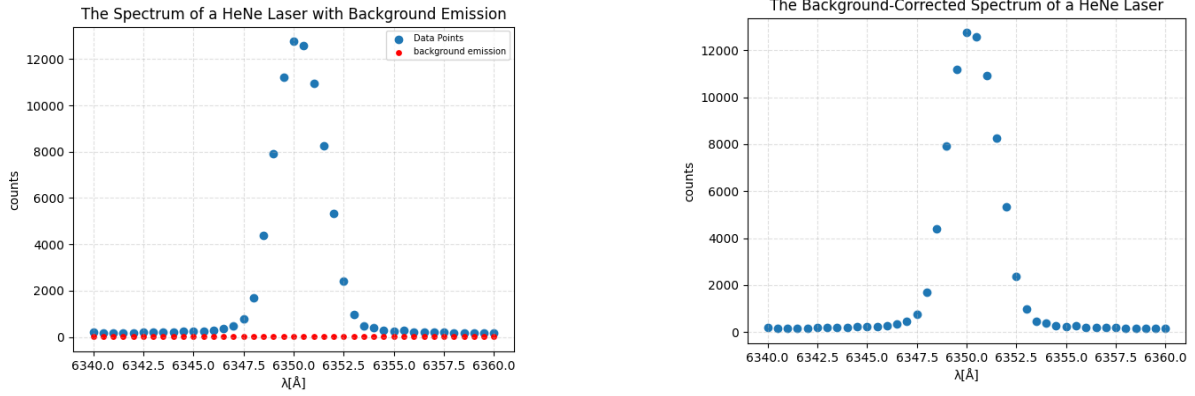


Figure 1: HeNe laser spectrum. (Left) Raw measurement including background signal. (Right) Background-corrected spectrum obtained by subtracting the background counts from the measured signal.

Figure 1 shows the measured spectrum of the *HeNe* laser before and after background subtraction. The left panel displays the raw spectrum together with the background signal, while the right panel shows the background-corrected spectrum. The correction was performed by subtracting the background counts at each wavelength from the measured signal:

$$Counts_{corrected} = Counts_{measured} - Counts_{background} \quad (4)$$

This procedure was applied consistently to all spectra, including the Balmer lines, to remove the influence of background noise.

Calibration Using *HeNe* Laser

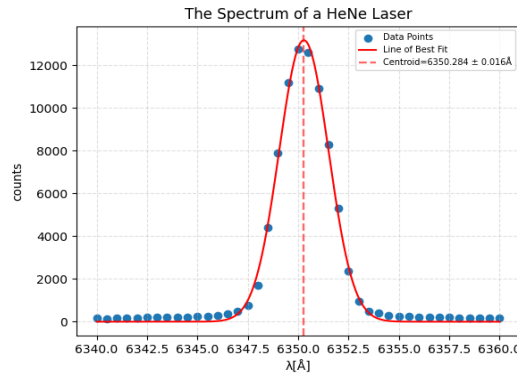


Figure 2: Background-corrected spectrum of the *HeNe* laser with Gaussian fit. The central wavelength from the fit was used to determine the spectrometer offset.

Figure 2 shows the background-corrected *HeNe* spectrum fitted with a Gaussian function. From the fit, the measured emission wavelength was determined to be:

$$\lambda_{measured} = 6350.27 \pm 0.016 \text{ \AA}.$$

The accepted value from the Bureau International des Poids et Mesures (BIPM, 2012) is:

$$\lambda_{HeNe, BIPM} = 6329.91 \pm 2.1 \times 10^{-11} \text{ \AA}.$$

The offset was therefore:

$$\lambda_{offset} = \lambda_{measured} - \lambda_{HeNe_{BIPM}} = 20.36 \pm 0.016 \text{ \AA}$$

This indicated that the spectrometer systematically overestimated wavelengths by this amount. **The calibration was applied to all Balmer measurements as:**

$$\lambda_{true} = \lambda_{measured} - \lambda_{offset} \quad (5)$$

The associated uncertainties were propagated using standard uncertainty analysis:

$$u(\lambda_{offset}) = \sqrt{[u(\lambda_{measured})]^2 + [u(\lambda_{true})]^2} \approx 0.016 \text{ \AA}$$

After calibration, the He–Ne wavelength was determined to be:

$$\lambda_{HeNe} = 6329.93 \pm 0.022 \text{ \AA}$$

This value agrees with the accepted BIPM value within experimental uncertainty.

Linearization of the Rydberg Constant

The Rydberg constant was determined using the linearized form of the Rydberg equation (2) (see Theory). where y represents the reciprocal wavelength and x is the theoretical term depending on n_i . A weighted linear least-squares fit was applied, accounting for uncertainties in the wavelength measurements.

The uncertainty of each reciprocal wavelength was calculated via error propagation:

$$u\left(\frac{1}{\lambda}\right) = \left| \frac{d}{d\lambda} \left(\frac{1}{\lambda} \right) \right| \cdot u(\lambda) = \frac{u(\lambda)}{\lambda^2} \quad (6)$$

These uncertainties were used to weight the fit, ensuring that more precise points carried greater influence in determining the slope, which corresponded to the Rydberg constant.

Results and Analysis

Balmer Series Spectrum

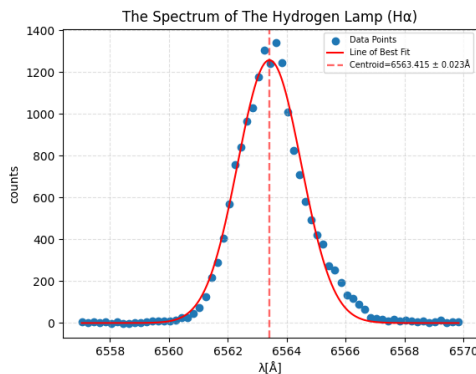


Figure 4: Measured spectrum of the H- α line with Gaussian fit

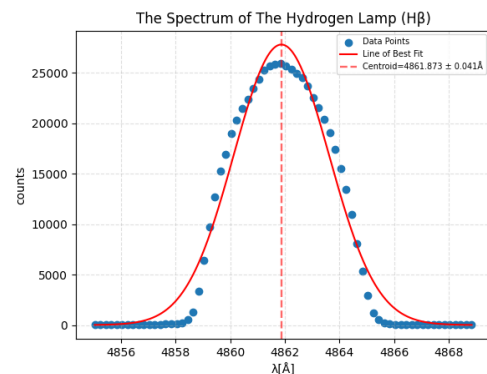


Figure 5: Measured spectrum of the H- β line with Gaussian fit.

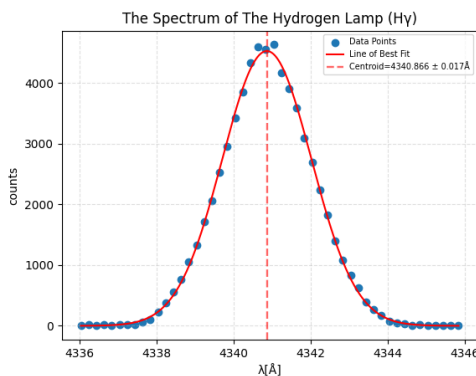


Figure 6: Measured spectrum of the H- γ line with Gaussian fit.

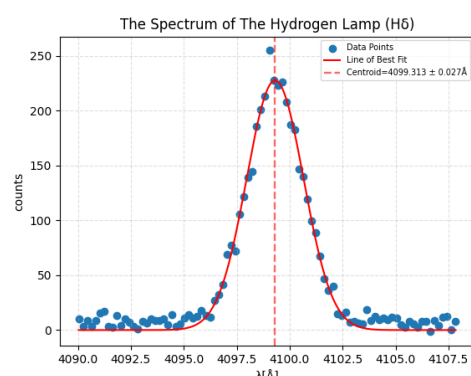


Figure 7: Measured spectrum of the H- δ line with Gaussian fit.

Analysis of Figures 4–7

The Gaussian function generally described the spectral data well, accurately modeling the peak shapes and central positions. However, in Figure 5 (H- β), the fit did not model the lower portions of the distribution as effectively as in the other cases. A more advanced function, such as a Voigt profile, could potentially improve the fit for this line.

The central emission wavelengths obtained from the Gaussian fits were:

$$\begin{aligned}\lambda_{H\alpha} &= 6563.42 \pm 0.023 \text{ \AA} \\ \lambda_{H\beta} &= 4861.86 \pm 0.041 \text{ \AA} \\ \lambda_{H\gamma} &= 4340.87 \pm 0.017 \text{ \AA} \\ \lambda_{H\delta} &= 4099.30 \pm 0.027 \text{ \AA}\end{aligned}$$

Determination of the Rydberg Constant

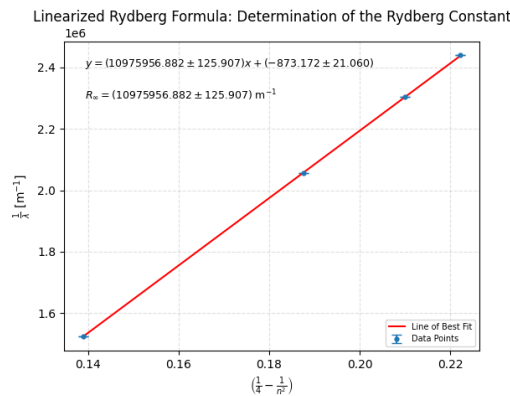


Figure 8: Linearized Rydberg plot. Reciprocal wavelengths of the Balmer lines are plotted against the theoretical term $(1/4 - 1/n_i^2)$. The slope of the best-fit line gives the experimental Rydberg constant.

Figure 8 shows the linearized Rydberg plot, constructed using the reciprocal wavelengths of the Balmer lines and their associated uncertainties. A weighted linear least-squares fit was performed, yielding the following equation for the line of best fit:

$$y = (10975956.87 \pm 125.91)x + (-873.16 \pm 21.05)$$

The gradient of the line corresponds to the Rydberg constant, giving:

$$R_H = 10975956.87 \pm 125.91 \text{ m}^{-1}$$

Analysis of Figure 8

The data points aligned closely with the fitted line, indicating strong agreement with the linearized Rydberg model. The error bars were very small, reflecting the high precision of the wavelength measurements.

Full Hydrogen Spectrum

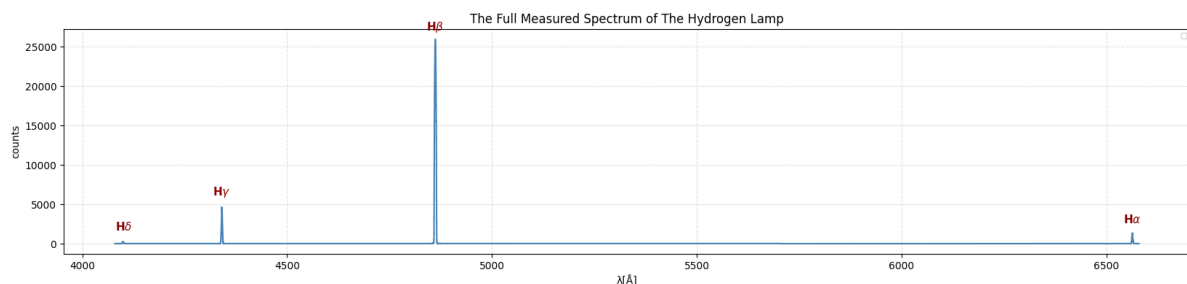


Figure 9: Full measured spectrum of the hydrogen lamp, showing the H- α , H- β , H- γ , and H- δ lines.

Figure 9 shows the combined spectrum of the hydrogen lamp, displaying all four Balmer lines.

Analysis of Figure 9

The measured intensities of the Balmer lines differed significantly. H- δ exhibited the lowest intensity, while H- β was the most intense. This observation raised two interesting considerations:

1. Whether the spectrometer had equal sensitivity across all wavelengths, and
2. Whether the hydrogen lamp emitted all Balmer lines with equal intensity.

These questions lie beyond the scope of this experiment but suggest possible avenues for further study.

Discussion

Determination of the Rydberg Constant

The experimentally determined value of the Rydberg constant was:

$$R_H = 10975956.87 \pm 125.91 \text{ m}^{-1}$$

The uncertainty of the measurement ($\pm 125.91 \text{ m}^{-1}$) was very small compared to the measured value, indicating that the experiment achieved a high degree of precision with low random error.

For comparison, the internationally recognized CODATA 2018 recommended value is:

$$R_{\text{accepted}} = 10967758.46 \pm 0.13 \text{ m}^{-1} \quad (\text{NIST, 2021})$$

The experimental result does not agree with the accepted value within experimental uncertainty. The measured constant is larger by 8198.41 m^{-1} . This discrepancy corresponded to a percentage error of approximately **0.075%**, which is small and demonstrates the overall accuracy of the method despite the lack of full agreement.

The deviation likely arose from systematic errors that were not fully accounted for. One major source might be the effect of the refractive index of air, since the theoretical Rydberg formula assumes measurements in a vacuum. In addition, the use of a Gaussian fit (particularly in the case of H- β , see Figure 5) may not have been sufficient to capture the true line shape. A Voigt profile, which incorporates both Gaussian and Lorentzian contributions, would likely improve the accuracy of line center determination.

Future improvements to this experiment could include correcting the measured wavelengths for the refractive index of air, applying more advanced fitting functions, and incorporating additional Balmer lines at shorter wavelengths to increase the dataset size and reduce statistical error.

Conclusion

The Balmer series spectrum of hydrogen was successfully measured, and the Rydberg constant was determined with high precision. While the result differed from the accepted CODATA value by 0.075%, the experiment demonstrated the reliability of spectroscopic methods for probing fundamental atomic constants. The small relative uncertainty confirmed the consistency of the measurements, and the identified discrepancies highlighted the importance of accounting for systematic effects.

Code and Data Availability

All datasets and results generated in this study, together with the Python scripts developed for calibration, uncertainty analysis, data analysis, and visualisation, are hosted in my GitHub repository: *Determination-of-the-Rydberg-Constant-from-Spectral-Data* (Lebepe, 2025).

References

1. Bureau International des Poids et Mesures. (2012). *Mise en pratique of the definition of the metre (2024)*. BIPM. Retrieved September 8, 2025, from https://www.bipm.org/documents/20126/41549578/M-e-P_unstab-HeNe_633.pdf/dcc689ee-8424-4cc8-ec76-1d33d187269b
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3. Lebepe, V. (2025). *Determination-of-the-Rydberg-Constant-from-Spectral-Data* [Code and Data repository]. GitHub. Retrieved September 8, 2025, from <https://github.com/ValenLebepe/Determination-of-the-Rydberg-Constant-from-Spectral-Data>